

Lab Submission [10m]

1. Create the network based on the diagram below by using RIP protocol and assign the IP address according to the table provided. **[1m]**



2. Display routing table on each router (R1, R2, and R3) by using command **show ip route**. Screenshot the results for R1, R2 and R3. **[6m]**

Screenshot (R1):

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

Router(config-if)#
Router(config-if)#enable
^
% Invalid input detected at '^' marker.

Router(config-if)#show ip route
^
% Invalid input detected at '^' marker.

Router(config-if)#enable
^
% Invalid input detected at '^' marker.

Router(config-if)#configure terminal
^
% Invalid input detected at '^' marker.

Router(config-if)#configure terminal
^
% Invalid input detected at '^' marker.

Router(config-if)#interface fastEthernet 0/0
Router(config-if)#ip address 10.0.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#ip address 10.0.2.1 255.255.255.252

Router(config-if)#interface fastEthernet 0/0
Router(config-if)#ip address 10.0.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#ip address 10.0.2.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#
Router(config-router)#no auto-summary
Router(config-router)#
Router(config-router)#network 10.0.1.0
Router(config-router)#
Router(config-router)#network 10.0.2.0
Router(config-router)#exit
```

Screenshot (R2):

```
Router(config-if)#
Router(config-if)#hostname R2
R2(config)#
R2(config)#interface serial 0/1/0
R2(config-if)#ip address 10.0.2.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface serial 0/1/1
R2(config-if)#ip address 10.0.3.1 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 10.0.2.0
R2(config-router)#network 10.0.3.0
R2(config-router)#exit
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
R       10.0.1.0/24 [120/1] via 10.0.2.1, 00:00:15, Serial0/1/0
C       10.0.2.0/30 is directly connected, Serial0/1/0
C       10.0.3.0/30 is directly connected, Serial0/1/1
R2#
```

Screenshot (R3):

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#interface serial 0/1/0
R3(config-if)#ip address 10.0.3.2 255.255.255.252
R3(config-if)#
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface fastEthernet 0/0
R3(config-if)#ip address 10.0.4.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#no auto-summary
R3(config-router)#network 10.0.3.0
R3(config-router)#network 10.0.4.0
R3(config-router)#exit
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
R       10.0.1.0/24 [120/2] via 10.0.3.1, 00:00:13, Serial0/1/0
R       10.0.2.0/30 [120/1] via 10.0.3.1, 00:00:13, Serial0/1/0
C       10.0.3.0/30 is directly connected, Serial0/1/0
C       10.0.4.0/24 is directly connected, FastEthernet0/0

R3#

```

3. Test all the connectivity to ensure the network is working. [3m]

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	4	(e...	(delete)
	Successful	PC0	PC3	ICMP		0.000	N	5	(e...	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	6	(e...	(delete)
	Successful	PC1	PC3	ICMP		0.000	N	7	(e...	(delete)

4. Save your file name as Lab Group_StudentID_StudentName_Lab3.docx.